Tutorial-1 (PH3102) Quantum Mechanics

① $$\left[-\frac{\hbar^2}{2m}\frac{\partial^2\psi}{\partial x^2} + V(x)\psi = i\hbar\frac{\partial\psi}{\partial t}\right] \quad \text{— (I)}$$

$$-\frac{\hbar^2}{2m}\frac{\partial^2\psi^*}{\partial x^2} + V(x)\psi^* = -i\hbar\frac{\partial\psi^*}{\partial t} \quad \text{— (II)}$$

$\psi^* \times$ (I) $-$ $\psi \times$ (II)

$$\Rightarrow -\frac{\hbar^2}{2m}\left[\psi^*\frac{\partial^2\psi}{\partial x^2} - \psi\frac{\partial^2\psi^*}{\partial x^2}\right] = i\hbar\left(\psi^*\frac{\partial\psi}{\partial t} + \psi\frac{\partial\psi^*}{\partial t}\right)$$

$$\Rightarrow -\frac{\hbar}{2im}\frac{\partial}{\partial x}\left(\psi^*\frac{\partial\psi}{\partial x} - \psi\frac{\partial\psi^*}{\partial x}\right) = \frac{\partial}{\partial t}(\psi\psi^*)$$

$$\Rightarrow \frac{\partial P}{\partial t} + \frac{\partial}{\partial x}\left[\frac{\hbar}{2im}\left(\psi^*\frac{\partial\psi}{\partial x} - \psi\frac{\partial\psi^*}{\partial x}\right)\right] = 0 \quad \text{— (1)}$$

$$\frac{d}{dt}\int_{x_1}^{x_2} P(x,t)\,dt$$

$$= \int_{x_1}^{x_2}\frac{\partial P}{\partial t}\,dt$$

$$= \int_{x_1}^{x_2} -\frac{\partial}{\partial x}\left[\frac{\hbar}{2im}\left(\psi^*\frac{\partial\psi}{\partial x} - \psi\frac{\partial\psi^*}{\partial x}\right)\right]dx \quad \text{(from (1))}$$

$$= \frac{\hbar}{2im}\left[\psi^*\frac{\partial\psi}{\partial x} - \psi\frac{\partial\psi^*}{\partial x}\right]_{x_2}^{x_1} = J(x_1,t) - J(x_2,t)$$

② For the ~~commutators~~ operators; A, B & C:

$$[A, BC] = [A,B]C + B[A,C]$$

$$\therefore [x, pe^{-p}] = [x,p]e^{-p} + p[x, e^{-p}]$$

$$= ie^{-p} + p[x, e^{-p}]$$

For $n \geq 1$:

$$[A, B^n] = [A, B\cdot B^{n-1}]$$

$$= [A,B]B^{n-1} + B[A, B^{n-1}]$$

$$= [A,B]B^{n-1} + B([A,B]B^{n-2} + B[A, B^{n-2}])$$

$$[A, B^n] = [A,B]B^{n-1} + B[A,B]B^{n-2} + B^2[A,B]B^{n-3} + \dots$$

Now, $[e^{-p}] = \sum_{n=0}^{\infty} \frac{(-1)^n}{n!} p^n$

$\Rightarrow [x, e^{-p}] = \sum_{n=0}^{\infty} \frac{(-1)^n}{n!} [x, p^n]$

Applying the same to $[x, p^n]$:

$[x, p^n] = \sum_{k=0}^{n-1} p^k [x, p] p^{n-1-k}$

$= \sum_{k=0}^{n-1} i p^k p^{n-1-k} = i n p^{n-1}$ (similar to taking derivative ~~are~~ ~~cab~~ $\frac{d}{dp} p^n = n p^{n-1}$)

$[x, e^{-p}] = \sum_{n=1}^{\infty} \frac{(-1)^n}{n!} \cdot i n p^{n-1}$

$= i \sum_{n=1}^{\infty} \frac{(-1)^n}{(n-1)!} p^{n-1}$

$= -i \sum_{n=1}^{\infty} \frac{(-1)^{n-1}}{(n-1)!} p^{n-1} = -ie^{-p}$.

$\therefore [x, pe^{-p}] = ie^{-p} - pie^{-p}$.

$= ie^{-p}(1-p)$ (Ans)